

VANG-Planner: Integrating Language Instructions into Spatial Cost for Global Path Planning

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Abstract—To operate in socially acceptable ways, robot navigation needs to consider various environmental and social constraints. Recent studies have explored learning-based methods and foundation models to capture such navigation preferences and incorporate them into path planning. However, in real-world deployments, preferred navigation behaviors often vary depending on spatial context and operational conditions. This makes it difficult to predefine all navigation rules in advance. As a result, many systems still rely on manual map editing. This paper proposes a global path planning framework that enables environment-specific navigation constraints to be incorporated after deployment. The proposed method constructs a Visual-Augmented Navigation Graph (VANG), a graph-based map augmented with robot-centric observation images stored on edges, and estimates instruction-dependent traversal weights. These weights are integrated into global planning costs without modifying the underlying map structure. Experiments in both simulation and real-world environments demonstrate that the proposed method handles diverse navigation instructions while maintaining competitive performance. The resulting paths are comparable to instruction-specific costmap baselines and close to human reference paths, without manual map modification.

Index Terms—Global Path Planning, Instruction-Aware Planning, Graph-Based Navigation, Language-Guided Navigation

I. INTRODUCTION

The ability of robots to select appropriate behaviors based on environmental context is essential for their acceptance in human society. Socially aware robot navigation has therefore been widely studied [1], as robots are required to exhibit preferred navigation behaviors [2]. To incorporate such environmental context into global path planning, layered costmaps and related methods encode additional navigation costs as separate layers [3, 4]. Another line of research explores language-guided robot navigation [5]. More recent works investigate language-guided navigation and reasoning with foundation models [6, 7, 8, 9, 10].

However, robots deployed in real-world environments need to account not only for general context but also for environment-specific navigation constraints. For example, in hospitals, peak congestion may vary across departments, while in supermarkets, pedestrian pathways can change between day and night. Such environment-specific constraints are difficult to anticipate, even for humans in unfamiliar environments. Existing costmap-based methods typically require the map to be defined before planning, making it

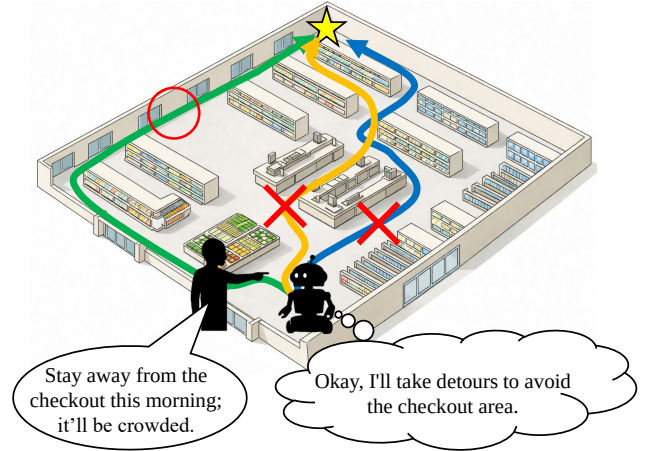


Fig. 1. Conceptual illustration of instruction-aware global path planning. After receiving an instruction (“Avoid the checkout area”), the robot consistently avoids the checkout area without manual map modification.

difficult to handle such constraints in advance. Methods using foundation models may incorporate such constraints through language instructions, but they mainly rely on the robot’s current observations, making it difficult to consider broader environmental context beyond the robot’s field of view.

In this work, we propose a global path planning framework that enables environment-specific navigation constraints to be incorporated after deployment. The key idea is to represent user-specified constraints as traversal weights derived from robot-centric observations and language instructions. Using natural language instructions and optional reference images, the proposed method enables instruction-aware global path planning without manual map editing.

The main contributions of this work are as follows:

- **Visual-Augmented Navigation Graph (VANG).** We introduce a graph-based map that stores robot-centric observation images on edges (edge-associated images), enabling environment-specific constraints to be encoded as local traversal weights.
- **Instruction-based traversal modeling.** We estimate instruction-dependent traversal weights from edge-associated images and language, allowing constraints to be incorporated without modifying the map.
- **Instruction-aware global planning on VANG.** We present a global planning framework that integrates estimated traversal weights into edge costs, enabling instruction-aware path planning under environment-specific constraints.

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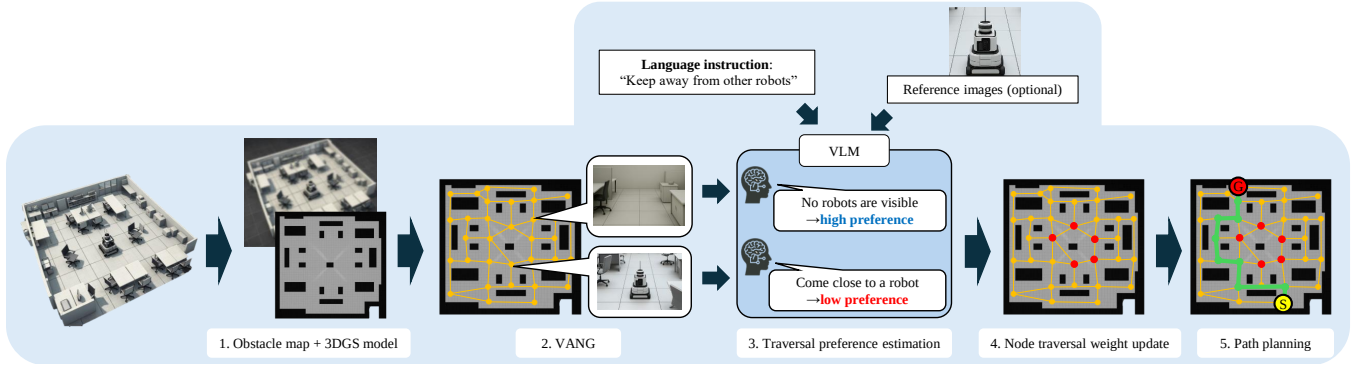


Fig. 2. Overview of the proposed VANG-Planner framework. VANG stores robot-centric observations associated with graph edges. A vision-language model outputs preference scoring based on language instructions and edge-associated images. The resulting scores are used to derive node traversal weights. The weights are incorporated into edge costs for instruction-aware global path planning.

II. PROBLEM FORMULATION

We consider a global path planning problem in which a robot moves from a start location to a goal location. Navigation preferences specified by language instructions are represented as traversal weights and incorporated into the planning cost. The traversable space of a two-dimensional obstacle map is modeled as a sparse graph $G = (V, E)$ with node set $V = \{v_1, \dots, v_n\}$ and directed edge set $E \subseteq V \times V$.

Given a language instruction L , each node v_i is associated with a traversal weight $w_L(v_i) > 0$.

Each edge e_{ij} has a geometric distance cost $d(e_{ij})$. The edge traversal cost is defined as

$$C(e_{ij}) = d(e_{ij}) f(w_L(v_j)), \quad (1)$$

where $f(\cdot)$ is a monotonic function that adjusts the cost according to the traversal weight, as described in Sec. III-C.

Given start and goal nodes v_s and v_g , a path $\pi = (v_{i_1}, \dots, v_{i_k})$ has total cost

$$C(\pi) = \sum_{t=1}^{k-1} C(e_{i_t i_{t+1}}). \quad (2)$$

The planned path is given by

$$\pi^* = \arg \min_{\pi \in \Pi(v_s, v_g)} C(\pi), \quad (3)$$

where $\Pi(v_s, v_g)$ denotes all feasible paths from v_s to v_g .

III. METHODOLOGY

An overview of the proposed framework is shown in Fig. 2. The method constructs VANG, a sparse navigation graph augmented with observation images (Sec. III-A). Traversal weights are estimated from language instructions and incorporated into edge costs for instruction-aware global planning.

A. Visual-Augmented Navigation Graph

To retain environmental context, we introduce the Visual-Augmented Navigation Graph (VANG) as a map representation for global path planning. VANG represents the traversable region as a sparse graph $G = (V, E)$. Robot-centric observation images I_{ij} are associated with each edge. Nodes represent reachable robot locations and edges represent traversable connections. The graph is initialized using a Voronoi-based decomposition, and the node density is subsequently adjusted. Specifically, in our implementation, the node spacing is set to approximately 1m considering the robot size and traversal width. Observation images are obtained using a 3D Gaussian Splatting pipeline [11] and associated with VANG edges.

Each edge is assigned a geometric distance cost $d(e_{ij})$, while each node is assigned a traversal weight $w_L(v_i)$. This weight reflects the suitability of traversing the region around the node under the given instruction. The traversal weights are estimated according to environment-specific constraints using the method described below.

B. Language-Guided Traversal Weight Estimation

Traversal weight estimation. The traversal weight $w_L(v_i)$ of each node is estimated from a language instruction L and observation images associated with edges entering node v_i . Let $\mathcal{E}^-(i)$ denote the set of such edges. For each edge $e_{ji} \in \mathcal{E}^-(i)$, the associated image I_{ji} is evaluated by a VLM with instruction L . The VLM outputs a preference score $s_{ji} \in \{1, \dots, K\}$ for traversing edge e_{ji} , where higher scores correspond to less preferred traversal. In this work, we set $K = 5$.

Preference scores from edges entering node v_i are aggregated to obtain a preliminary weight

$$\tilde{w}_L(v_i) = \frac{1}{|\mathcal{E}^-(i)|} \sum_{e_{ji} \in \mathcal{E}^-(i)} s_{ji}.$$

Because scores differ across observation directions, smoothing is applied over neighboring nodes. Let $\mathcal{N}(i)$

Role: You are a robot navigating inside a building.
Navigation condition (variable): <i>[task-specific instruction]</i> (e.g., Do not pass near rectangular cardboard boxes.)
Evaluation task: Evaluate the cost of moving forward 1 meter on a 5-level scale.
Constraints: • Collision possibility is ignored. • Output 3 if the image is invalid.
Output format: Answer: 1 2 3 4 5 Reason: [brief explanation]

Fig. 3. **Language Prompt for Traversability Scoring.** The prompt follows a shared structure across all scenarios; only the navigation-condition field (in italics) is modified to specify task-dependent navigation behavior.

denote nodes connected to v_i . The final weight is

$$w_L(v_i) = (1 - \lambda)\tilde{w}_L(v_i) + \lambda \frac{1}{|\mathcal{N}(i)|} \sum_{v_j \in \mathcal{N}(i)} \tilde{w}_L(v_j), \quad (4)$$

where $\lambda \in [0, 1]$ is a smoothing parameter. In this work, we set $\lambda = 0.5$.

Prompt design. An example prompt is shown in Fig. 3. The prompt consists of (i) a navigation condition and (ii) a fixed evaluation and output format. Across scenarios, only the navigation condition is changed. To handle instructions difficult to express using language alone, reference images can optionally be provided and described in the prompt. We employ GPT-5.2 via the OpenAI API with default sampling parameters. The model outputs a preference score from 1 to 5 for each observation image. We also examine the effect of the VLM choice (see Sec. VI-B).

C. Path Planning on VANG

Path planning searches for the minimum-cost path between a start and goal node using edge costs on VANG. The edge traversal cost $C(e_{ij})$ is defined by scaling the geometric distance $d(e_{ij})$ using the traversal weight of the destination node v_j . The weight $w_L(v_j) \in [w_{\min}, w_{\max}]$ is normalized to the range $[-1, 1]$ as

$$z(v_j) = \frac{2(w_L(v_j) - \frac{w_{\min} + w_{\max}}{2})}{w_{\max} - w_{\min}}.$$

A parameter ρ controls the influence of the instruction, and the scaling function is defined as

$$f(w_L(v_j)) = \exp((\ln \rho) z(v_j)). \quad (5)$$

The final edge traversal cost is given by

$$C(e_{ij}) = d(e_{ij}) f(w_L(v_j)). \quad (6)$$

In this work, we set $\rho = 6$.

Paths on VANG consist of straight segments between nodes. We refine the VANG path into a smooth executable path by local grid-based A* search within a 1.3 m-wide area around the VANG path, considering the robot footprint and safety margins.

IV. EXPERIMENTS

A. Experimental Setup

Simulation environment. To evaluate global path planning over a wide area, we use an Unreal Engine simulation representing a floor of Toyota Memorial Hospital. An area of approximately $100 \text{ m} \times 20 \text{ m}$ is used for evaluation. Data are collected using a robot equipped with an RGB camera (640×480 px, FoV 120°) and a LiDAR sensor for VANG construction.

Real-world environment. For real-world evaluation, we use (i) an office environment and (ii) a real hospital environment (Toyota Memorial Hospital). The office and hospital areas are approximately $15 \text{ m} \times 10 \text{ m}$ and $20 \text{ m} \times 15 \text{ m}$, respectively. Data are collected using Toyota's Human Support Robot [12], equipped with an RGB camera (4416×1242 px, FoV 120°), to construct a two-dimensional obstacle map and a 3DGS model. The 3DGS model is built from camera data collected during approximately 10 minutes of robot traversal in each environment. VANG is constructed from these data.

B. Baselines

Costmap-based approaches are closely related to our method, but their costmaps typically encode general semantic information for planning [4, 13]. Recent VLM-based navigation methods often rely on online observations for local decision making [8, 9], whereas VANG-Planner uses map-stored observations for global path planning; these roles are complementary. Given this scope, we include an instruction-specific costmap baseline, where target objects are manually annotated for each instruction and incorporated into planning cost. In addition, to evaluate the plausibility of the generated paths, we include a human reference path created by a robot operator with sufficient knowledge of the environment and operational constraints.

Grid-based A*. A* search [14] computes the shortest path on an occupancy grid.

Instruction-Specific Costmap (ISC). A heuristic baseline in which navigation constraints are manually encoded into a costmap prior to planning. Traversal costs are increased within a fixed radius (1.5 m) around objects annotated for each instruction. This baseline assumes that environment-specific constraints are known in advance. Unlike the proposed method, the map requires manual modification for each instruction.

Ours without Node Traversal Weight (w/o NTW). An ablation variant of the proposed method in which the node traversal weight $w_L(v_i)$ is removed from the cost computation.

Human-guided planning (reference). A global path manually created by a robot operator is used as a reference. The operator was not involved in the annotation process described in Sec. VI-A and had no access to the annotation results. This path serves as a reference for practical human navigation behavior under the given operational constraints.

TABLE I

SCENARIO CONFIGURATIONS AND CORRESPONDING PROMPTS. ALL SCENARIOS USE THE PROMPT TEMPLATE SHOWN IN FIG. 3. EACH SCENARIO SPECIFIES A NAVIGATION CONDITION; +REF ROWS INCLUDE ADDITIONAL INSTRUCTIONS FOR INTERPRETING REFERENCE IMAGES.

Scenario	Navigation condition (variable field)	Inputs
S1	Avoid areas near facilities directly used by people (e.g., sofas or medical equipment) whenever possible.	Query
S1 +Ref	<i>Additional instruction:</i> Use Ref. 1–3 to calibrate distance (example object: sofa; near / moderate / far).	Ref. 1–3 + Query
S2	Prefer moving on dark-colored floor mats whenever possible.	Query
S2 +Ref	<i>Additional instruction:</i> Ref. 1 shows a recommended area, and Ref. 2 shows a non-recommended area.	Ref. 1–2 + Query
S3	Avoid areas near outpatient department N (crowded) and prefer areas near department E (less crowded).	Query
RO4	Do not pass near rectangular cardboard boxes.	Query
RO5	Whenever possible, take the route between the monitor and the bookshelf.	Query
RH6	Do not pass by a vending machine.	Query

TABLE II

VIOLATION RATIO (%) ↓ ON PAAs. THE VIOLATION RATIO DENOTES THE PERCENTAGE OF THE PLANNED PATH OVERLAPPING PAAs. LOWER VALUES INDICATE BETTER COMPLIANCE WITH AVOIDANCE INSTRUCTIONS. RESULTS ARE REPORTED FOR THREE CONSENSUS LEVELS ($k = 3, 5, 8$). ISC DENOTES THE INSTRUCTION-SPECIFIC COSTMAP BASELINE. BOLD VALUES INDICATE THE BEST RESULT AMONG ALGORITHMIC METHODS.

Method	S1			S3			RO4		
	$k = 3$	$k = 5$	$k = 8$	$k = 3$	$k = 5$	$k = 8$	$k = 3$	$k = 5$	$k = 8$
Grid-based A*	27.6%	22.7%	21.1%	23.4%	22.6%	12.6%	17.6%	15.2%	5.8%
ISC	16.8%	9.4%	0.0%	–	–	–	8.4%	0.0%	0.0%
Ours w/o NTW	14.8%	16.4%	13.9%	23.4%	22.2%	14.9%	14.4%	4.9%	0.0%
Ours	11.4%	0.4%	0.0%	1.7%	0.0%	0.0%	0.0%	0.0%	0.0%
Ours (+Ref)	7.8%	2.4%	0.0%	–	–	–	–	–	–
Human-guided	3.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

TABLE III

TRAVERSAL RATIO (%) ↑ ON PTAs. THE TRAVERSAL RATIO IS THE PERCENTAGE OF THE PLANNED PATH OVERLAPPING PTAs. HIGHER VALUES INDICATE BETTER COMPLIANCE WITH TRAVERSAL INSTRUCTIONS. FOR S3 AND RO5, RESULTS ARE REPORTED FOR THREE CONSENSUS LEVELS ($k = 3, 5, 8$).

Method	S2	S3			RO5		
	fixed PTA	$k = 3$	$k = 5$	$k = 8$	$k = 3$	$k = 5$	$k = 8$
Grid-based A*	62.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Ours w/o NTW	84.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Ours	76.2%	22.1%	14.8%	6.1%	20.5%	18.7%	15.9%
Ours (+Ref)	96.8%	–	–	–	–	–	–
Human-guided	100.0%	19.9%	14.4%	5.9%	21.0%	19.7%	16.1%

C. Scenarios

We categorize navigation instructions into two types: Preferred Avoidance Areas (PAAs), i.e., regions the robot should avoid, and Preferred Traversal Areas (PTAs), i.e., regions the robot is encouraged to traverse. We define six scenarios: three in simulation (S1–S3), two in a real office (RO4, RO5), and one in a real hospital (RH6).

- **S1:** PAA scenario for avoiding facilities (e.g., sofas).
- **S2:** PTA scenario for designated walking zones.
- **S3:** Combined PAA/PTA scenario under department-wise congestion conditions.
- **RO4:** PAA scenario for avoiding cardboard boxes.
- **RO5:** PTA scenario for traversing the space between the bookshelf and the monitor.
- **RH6:** PAA scenario for avoiding a vending machine.

PAAs and PTAs for S1, S3, RO4, and RO5 are determined from annotation data collected in a human study (see Sec. VI-A). For S2, the PTA is uniquely determined by the environment. RH6 shares the same task structure as RO4 (PAA scenario); therefore, RH6 is evaluated qualitatively only.

Floor color differences in S2 and appropriate object distances in S1 are difficult to express using text alone. To address this, reference images are provided (three in S1 and two in S2). We also compare performance with and without reference images.

Start and goal positions are defined for each scenario so that the robot travels sufficiently long distances and encounters multiple route options. All methods are evaluated under identical start and goal configurations. Scenario instructions and reference images are summarized in Table I.

D. Overall Evaluation

Metrics. To evaluate how well planned paths follow the intended navigation conditions, we measure how much of each planned path lies inside PAAs and PTAs. PAAs and PTAs are defined based on human annotations collected from 10 participants. For each location, we count how many participants selected the region and define a consensus area using a threshold k . We use three consensus levels ($k = 3, 5, 8$) in the evaluation. Details of the annotation procedure are provided in Sec. VI-A.

Let L_{plan} denote the total length of a planned path, and L_{PAA} the path length inside PAAs. The violation ratio is defined as

$$\text{Violation ratio} = \frac{L_{\text{PAA}}}{L_{\text{plan}}}. \quad (7)$$

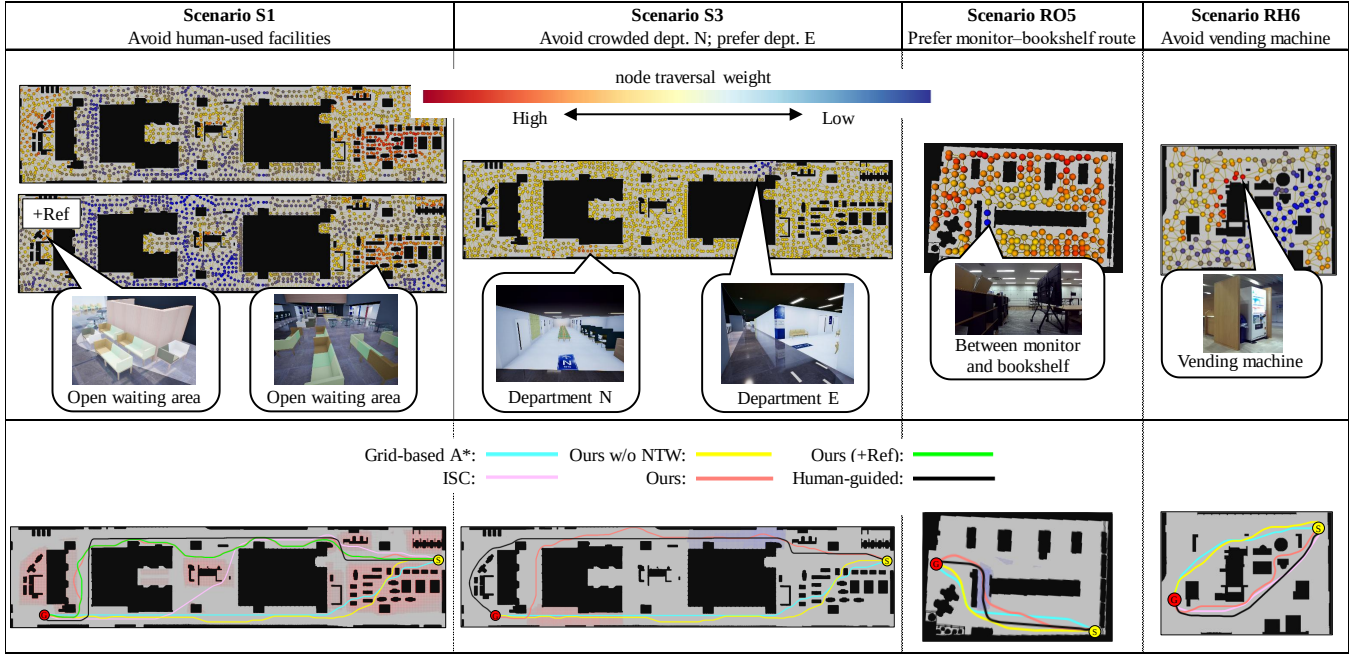


Fig. 4. Qualitative visualization across scenarios. Each column corresponds to a scenario, with the instruction summary shown at the top. **Second row:** VANG after node traversal weight updates. Nodes are colored by traversal weight (blue: low, yellow: medium, red: high). Instruction regions are highlighted. **Third row:** Planned paths. Red and blue shaded regions denote PAAs and PTAs, respectively, with shading indicating human consensus ($k=3,5,8$).

A smaller value indicates better compliance with avoidance condition.

Similarly, let L_{PTA} denote the path length inside PTAs. The traversal ratio is defined as

$$\text{Traversal ratio} = \frac{L_{PTA}}{L_{plan}}. \quad (8)$$

A larger value indicates better compliance with traversal condition.

Quantitative Results. Table II and III report violation and traversal ratios for the evaluated scenarios. ISC is applied only when target object categories can be directly specified from the instructions (S1 and RO4). Overall, the proposed method improves compliance over Grid-based A* and achieves strong performance across scenarios, approaching the human-guided reference.

Compared with Grid-based A*, the proposed method produces longer paths (up to 22.9%). Human-guided reference paths are also longer, with increases of up to 21.5%.

Qualitative Results. Figure 4 shows planned paths and VANG representations for scenarios S1, S3, RO5, and RH6. When multiple route options exist, the proposed method selects routes closer to the human-guided reference.

In S3 and RO5, nodes near PAAs show higher traversal weights, while those near PTAs show lower weights. In S1, without reference images, regions where the target object is visible tend to have uniformly high weights. In RH6, some regions beyond the vending machine also show high weights.

Discussion and Limitations. Tables II and III show that the proposed method achieves high compliance with both avoidance and traversal instructions across all scenarios. Quanti-

tative results exhibit trends comparable to the human-guided reference, with path length increases remaining within a similar range. Qualitative results further indicate that, when multiple route choices are available, the proposed method tends to select paths similar to those chosen by humans.

The focus of this study is to enable environment-specific constraints to be incorporated without manual map editing. While the Instruction-Specific Costmap (ISC) baseline requires per-scenario costmap modification, the proposed method achieves comparable performance. This demonstrates the feasibility of path planning under environment-specific constraints without manual map modification.

Clear improvements are observed in scenarios with explicitly defined instructions (S2, S3, RO4, and RO5). Even for abstract instructions such as S1, the generated paths remain close to the human-guided reference. This suggests that the method can reflect implicit spatial intent beyond explicitly defined constraints. Furthermore, reference images tend to produce more spatially coherent weight distributions. This suggests that visual references help interpret instructions that are difficult to express in language alone.

In RH6, however, high traversal weights were also assigned to objects visually similar to vending machines. This behavior reflects misclassification caused by local visual similarity and highlights the difficulty of semantic interpretation based solely on local observations. Humans can often resolve such ambiguity by considering broader spatial context. As discussed in Sec. VI-C, leveraging multiple images stored in VANG may help resolve this issue by providing broader spatial context. Exploring such context-aware comparison within VANG is an important direction for future work.

V. CONCLUSION AND FUTURE WORK

This paper presented VANG-Planner, a framework for incorporating environment-specific navigation constraints into instruction-aware global path planning after deployment. The method estimates traversal weights from language instructions and local observations and integrates them into the planning cost, enabling adaptive navigation without manual map modification.

Experiments in simulation and real-world environments show that VANG-Planner handles diverse instructions and generates paths consistent with the intended navigation constraints. The resulting paths are comparable to instruction-specific costmap planning and close to human-guided reference paths. Reference images further help when spatial intent is difficult to express in language alone.

Limitations may remain in cases involving visually similar objects or decisions requiring broader spatial context, where reliance on local observations alone may make traversal weight estimation unstable. Future work will explore wider spatial context, structural reasoning, and closed-loop execution with a local controller for more robust instruction-aware navigation.

VI. ADDITIONAL EXPERIMENTS AND DETAILS

This section provides additional experimental analyses and details of the evaluation protocol used in the main experiments.

A. Human Annotation of PAAs and PTAs

To evaluate how well planned paths follow the intended navigation conditions, PAAs and PTAs for S1, S3, RO4, and RO5 are defined based on human annotations.

Annotation data collection. We conducted a user study with 10 participants. Each participant annotated PAA and PTA regions as polygons on a 2D map according to the navigation condition. Informed consent was obtained from all participants, and the study was conducted in accordance with the “Ethical Guidelines for Research” of Toyota Motor Corporation.

Annotation analysis. To define PAAs and PTAs used for evaluation, we analyze the annotations using an agreement metric. Let A_i denote the region annotated by participant i for a given area type (PAA or PTA). For each participant i , we compute the Intersection over Union (IoU) between A_i and the regions annotated by all other participants, and define the agreement score s_i as the median IoU:

$$s_i = \text{median}_{j \neq i} \text{IoU}(A_i, A_j). \quad (9)$$

Agreement scores vary across scenarios, typically ranging from 0.2 to 0.4. An overlay of annotations for S3 is shown in Fig. 5. PTAs exhibit larger variability than PAAs, but regions selected by multiple participants still reveal shared spatial tendencies.

We define a *consensus area* as a region selected by at least k participants. For each location x on the map,

$$C_k = \{x \mid \sum_i \mathbf{1}(x \in A_i) \geq k\}. \quad (10)$$



Fig. 5. Overlay visualization of PAA and PTA annotations from all participants (Scenario S3). Red regions indicate PAAs, while blue regions indicate PTAs.



Fig. 6. Without a reference image (top-left), the payment machine (bottom-right) is misidentified as a vending machine (bottom-left), leading to high traversal weights. With a reference image (top-right), high weights are not assigned around the payment machine.

Three consensus levels ($k = 3, 5, 8$) are used as reference regions in the evaluation.

B. Effect of VLM Choice

To examine the effect of the VLM choice, we compared the scores assigned by different OpenAI models to the same set of images. In scenario RO5, all images stored in VANG (830 images) were scored using GPT-4o and GPT-o3, while the scores from GPT-5.2 were used as the reference. All models were used with default API settings.

Let $s_{\text{ref}}(i)$ denote the score produced by GPT-5.2 for image i , and $s_m(i)$ the score assigned by another model m . We measure the ratio of images satisfying $|s_m(i) - s_{\text{ref}}(i)| \leq 1$ over all images, which indicates how often two models produce similar scores.

Across all images, the ratios were 0.796 for GPT-4o and 0.758 for GPT-o3, while a random baseline achieved 0.583. These results suggest that the scoring tendency is largely consistent across different VLM choices.

C. Reference Images for Misclassification Cases

As shown in Fig. 6, in RH6, high traversal weights were incorrectly assigned to the payment machine due to visual similarity with vending machines. Providing a reference image of the target vending machine reduced this misclassification. Since VANG stores images across the environment, comparing multiple images may help mitigate such errors caused by local visual ambiguity.

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